

Hw

Q. Find the frequency counts for the following.

i) for ($i=1; i < n; i++$)
 printf("Hello");

ii) $i=1$
 $i = i + 20;$
 printf("%d", i);

iii) $i=1;$
 while ($i <= n$)
 {
 printf("%d", i);
 $i++$;
 }

Q2. For all the shell programs done in OS lab - Find out the time complexity using frequency count

Ans: i)

	Cost	repetition	total
$i=1$	1	1	1
$i <= n$	1	n	n
$i++$	1	n	n
printf("Hello");	1	1	1
			<u>$2n+2$</u>

ii)

```
i = 1
i = i + 20
print("i.d", i)
```

cost	repetition	total
1	1	1
2	1	2
1	1	1
		<u>4</u>

iii)

```
i = 1
while(i <= n){
    print("i.d", i);
    i++;
}
```

cost	repetition	total
1	1	1
1	n+1	n+1
1	n	n
1	n	n

3n+2

3n+2

Ans 2.

Ans2: i) # !/bin/bash

echo "Hellowor ld" ⇒ 1

ii) echo "Enter num1" ⇒ 1

read num1 ⇒ 1

~~echo num1~~

echo "Enter num2" ⇒ 1

read num2 ⇒ 1

sum=\$((num1+num2)) ⇒ 1

echo "Sum of \$num1 and \$num2 is = \$sum" ⇒ 1

$$\begin{aligned} \text{Total} &= 1+1+1+1 \\ &\quad +1+1 \\ &= \underline{\underline{6}} \end{aligned}$$